Practical No : 3

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Q.3) Write a C program for Quick Sort.

```
#include <stdio.h>
#include <conio.h>
void quickSort(int arr[], int low, int high)
{
   int i = low, j = high;
   int pivot = arr[(low + high) / 2];
   int temp;
 while (i <= j)
   {
      while (arr[i] < pivot)
         i++;
 while (arr[j] > pivot)
         j--;
  if (i <= j)
      {
         temp = arr[i];
         arr[i] = arr[j];
         arr[j] = temp;
         i++;
         j--;
      }
   }
  if (low < j)
    quickSort(arr, low, j);
 if (i < high)
      quickSort(arr, i, high);
```

```
}
int main() {
    int arr[100], n, i;
   printf("Enter the number of elements: ");
    scanf("%d", &n);
  printf("Enter the elements:\n");
    for (i = 0; i < n; i++)
        scanf("%d", &arr[i]);
 quickSort(arr, 0, n - 1);
 printf("Sorted array:\n");
    for (i = 0; i < n; i++)
        printf("%d ", arr[i]);
  return 0;
}
```

Output:

